

GeoSentinel AI — Analysis Report

Generated: 02 July 2026, 00:18

Analysis Period

Field	Value
Date 1 (T1)	2023-01-14
Date 2 (T2)	2024-01-14
Study Area	Hyderabad Metropolitan Region (HMR)
Satellite	Sentinel-2 L2A
Resolution	10 metres

Land Cover Area Change

Class	T1 Area (km²)	T2 Area (km²)	Change (km²)	Change (%)
Background	53.749	65.989	+12.240	+22.8%
Urban	0.000	0.000	+0.000	+0.0%
Vegetation	25.462	13.419	-12.042	-47.3%
Water	1.070	0.868	-0.202	-18.9%
Barren	0.009	0.014	+0.005	+50.5%
Agriculture	0.000	0.000	+0.000	+0.0%

Recommendations

[CRITICAL] Critical Vegetation Loss Detected

Why: Vegetation cover decreased by 39.3% (30.45 km²) between 2023-01-14 and 2024-01-14, which exceeds the critical threshold of 25%.

Action: A critical level of vegetation cover has been lost in the selected area. Immediate field investigation is recommended. Initiate emergency afforestation programs and restrict any further land clearing activities. Notify the Telangana Forest Department for intervention.

[HIGH] Significant Vegetation Loss Detected

Why: Vegetation cover decreased by 39.3% (30.45 km²) between 2023-01-14 and 2024-01-14, exceeding the high-alert threshold of 15%.

Action: Significant vegetation loss has been detected. Plan and execute targeted afforestation programs in affected zones. Identify drivers (construction, agriculture conversion, fire) through ground truth verification. Monitor monthly for further changes.

[HIGH] Land Cover Change Hotspots Identified

Why: 236 change hotspot(s) were detected, each exceeding 0.5 km². The largest hotspot covers 1.16 km² centered at 17.3850°N, 78.4867°E.

Action: Concentrated change hotspots have been identified. These may indicate targeted development activity or environmental disturbance. Dispatch field teams to highest-priority locations for immediate ground verification and documentation.